



KTU NOTES

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**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPER**

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midpoint and bridge configuration if the firing angle is 50° and the rating (peak) of the thyristors is 200V.

5. A single phase semi-converter fed from 120 V, 50 Hz supply is connected to a load with resistance of $10\ \Omega$. If the average output voltage is 25% of its maximum, find the average output voltage, find the circuit turn off time.
6. A three-phase half-wave controlled converter is connected to 380 V, 50 Hz supply. The load current is constant at 32 A independent of the firing angle α . The forward drop of SCRs is 1.2 V, Find: i) Peak reverse voltage rating, ii) Average power dissipation in each SCR
7. A single-phase half bridge inverter has a resistive load of $10\ \Omega$, and an input voltage of 96 V. Obtain the Fourier series representation of the output voltage waveform and hence find the value of distortion factor.
8. A three phase half wave converter is operated from 3-phase, 230 V, 50 Hz supply. The load resistance $R = 10\ \Omega$. An average output voltage of 50% of the maximum average output voltage is required. Determine i) the firing angle, ii) average output current.
9. Explain the operation of three-phase dual converter with circulating current.
10. Describe a single phase half controlled converter with RL load along with its circuit diagram and waveforms
11. Draw the circuit and derive the expression for output voltage of a single phase half controlled converter.
12. Sketch the waveform of input voltage, output voltage and output current for a single phase fully controlled converter with R load with $\alpha = 0^\circ$
13. Draw the output voltage waveform of a 3-phase controlled half wave rectifier with $\alpha = 30^\circ$.
14. Obtain an expression for average dc output voltage of a 1-phase full bridge rectifier for R load with firing angle, α .
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Part B

converter with RL load. Draw the waveform of output voltage with freewheeling diode and output current.

7. With the help of circuit diagram explain the operation of single phase half controlled converter with RL load. Draw the waveform of input voltage, output voltage, voltage across the thyristor
8. Explain the operation of three-phase dual converter with circulating current
9. Describe a single phase half controlled converter with RL load along with circuit diagram and waveforms.
10. A fully controlled full wave converter has a source of 240 V rms, 50mH, 50V Emf opposing series load. The delay angle is 45° . Determine a) Average output voltage and current. b) Rms load voltage and Rms voltage across the load. c) The power absorbed by the 50V load back emf
11. Illustrate how a Thyristor based 1-phase fully controlled rectifier can convert ac into variable dc. Draw the waveforms of output voltage and current for both R and RL load at $\alpha=30^\circ$.
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